

Forensic Report: The "Epoch Reset" Bug

Date: June 23, 2026

Subject: Analysis of Model Training Failure and Checkpoint Corruption

Executive Summary

An analysis of the `trainer_state.json` logs has identified the root cause of the model training failure. The catastrophic loss of model performance and the subsequent NaN crash were caused by altering the `num_train_epochs` parameter during the resume process. This action reset the Learning Rate (LR) schedule, causing destabilizing spikes in the gradient and weight updates.

1. Root Cause Analysis: The Learning Rate Reset

The primary failure occurred when the model resumed from a checkpoint, during which the `num_train_epochs` value was adjusted.

- **Observation (Step 1000 to 1010):**
 - **Step 1000 (Pre-resume):** `learning_rate` was at a stable 0.000025, with a healthy loss of 4.91.
 - **Step 1010 (Post-resume, epochs changed 1 → 3):** The `learning_rate` spiked to 0.000171.
- **Impact:** Modifying `num_train_epochs` or `max_steps` during resumption forces the Hugging Face Trainer to discard the existing LR schedule. A new warmup/decay

cycle was initiated, subjecting the model to a maximum learning rate that obliterated the previously learned phonetic alignments.

2. Cascading Failure: The Step 3500 Restart

A secondary adjustment at step 3500 (changing epochs to 4) introduced a third wave of maximum learning rate shocks.

- **Mechanism of Failure:** By resuming without gradient clipping (`max_grad_norm`), the learning rate shocks caused gradients to grow exponentially.
 - **Result:** By step 4600, the gradients exceeded numerical stability limits, triggering the NaN (Not a Number) crash.
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Recommendations for Future Runs

The updated environment now includes `max_grad_norm = 0.3`, `fp16 = True`, and `num_train_epochs = 10` (Note: 10-epoch safety net to prevent LR resets) to provide protection against exploding gradients and training instability. To prevent the recurrence of the LR reset bug, strict adherence to the following procedure is required:

CRITICAL: Never change `num_train_epochs` when resuming.

- Define the total number of epochs (e.g., `num_train_epochs = 10`) at Step 0 before the initial training run.
- Upon resuming from a timeout, maintain the original value. The Trainer will correctly identify the completed progress and resume the remaining training cycles without resetting the learning rate schedule.